Development of an Optical Tracking System for Real-time Robotic Applications

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Abstract

Engineering students often need a clear, practical model for writing dissertations; this document provides an overview tailored to final-year mechatronics projects and demonstrates effective use of a LaTeX thesis template. Many student reports lack consistent structure, rigorous scoping, and appropriate use of figures, tables, code listings, and citations, which impedes readability and assessment.

This work contributes an exemplar that explains what each report section should contain, offers typical subheadings, and embeds working examples of template elements to copy and adapt. The document is implemented in XeLaTeX using the `memoir` class with chapters for Introduction, Literature Review, Methodology, Results, Discussion, Conclusions, Recommendations, and Appendices, along with acronyms and bibliography support; the result is a shareable guide that doubles as a starting point for students to produce professional, evidence-based engineering reports.

How to write an abstract in 5 minutes?

Answer the following questions in narrative form with no more than two sentences per question:

- *What is the background to the project?*
- *What is the problem that is being solved?*
- *What is the contribution of the project to this problem?*
- *What is the main conclusion of the project?*

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AGP Art Gallery Problem 7

GDOP Geometric Dilution of Precision 7

MoCap Motion Capture 9

OCP Optimal Camera Placement 7

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Resolution, latency, iframe rate, microcontroller...

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flow chart

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5.1 Literature Review

The effectiveness of an indoor localisation system depends on the strategic positioning of the sensor network. The application influences what the distribution prioritises. For example, it may be necessary to ensure the entire area is covered with sufficiently distinct viewpoints to capture elaborate motion, or coverage serves as a mean to be able to view the entire expanse, or it may need to be configured to mitigate the Geometric Dilution of Precision (GDOP) arising from the sensor network's limited resolution [1]. This placement problem constitutes an optimisation task involving determination of both the quantity and pose of sensors so as to improve measurement quality by not exacerbating uncertainty due to poor placement. This section explores the various approaches to the Optimal Camera Placement (OCP) problem; examining both the (1) characterisation and modeling of sources of inherent uncertainty, attributable to image quantisation and occlusion of the target, and strategies for addressing them, as well as (2) the range of optimisation methods employed, from heuristic algorithms to integer programming.

5.1.1 Background of OCP

The OCP descends from the well-studied Art Gallery Problem (AGP), which investigated the minimum number of guards required to observe an entire 2D polygon with n vertices. The AGP assumed guards had unlimited 360° field of view, infinite range, and operated in two dimensions only. Fisk elegantly proved that $\lfloor \frac{n}{3} \rfloor$ guards are always sufficient to cover any simple polygon, using a proof based on triangulation and 3-colouring; in practice, specific polygons often require fewer, as demonstrated in Figure In this example, a polygon with $n = 13$ vertices is first triangulated, then each vertex is assigned one of three colours such that no triangle shares two vertices of the same colour; placing guards at all vertices of the least frequent colour guarantees complete coverage. Similar to the AGP the OCP must take a heauristic goal and mathematically model it in pursuit that given the amount of sensors the arrangment. The optimal camera placement problem extends this foundation by introducing constraints that reflect the true capabilities and limitations of physical sensors. Moreover, the OCP seeks to extend beyond mere coverage as in the case of accurate triangulation more than one camera needs to view a point in the target area.

5.1.2 Motivation for Optimal Placement

These constraints are motivated by diverse applications, such as: surveillance and monitoring [2], motion capture [3, 4], augmented and virtual reality [5], and autonom-

ous driving [6]. In such visual sensor networks, accurate localisation requires that each marker or feature be visible to at least two cameras, enabling 3D position recovery through triangulation [3, 4]. Most papers emphasise that sensor placement (including amount, location, orientation, and arrangement to the other sensors in the network) significantly affects system coverage and tracking performance [7]. However, the OCP is NP-hard and combinatorial; this means that the search space grows explosively with the number of candidate positions and orientations, rendering exhaustive search intractable. Consequently, most investigators resort to heuristic-based approximation algorithms as they offer a practical path to reach acceptable solutions within reasonable computation time.

Generally, the OCP can be stated as follows: given a set of candidate camera locations and orientations, find the configuration that maximises 3D reconstruction quality of a target within a defined workspace, subject to practical constraints such as mounting options and budget. Solving this problem requires balancing trade-offs among coverage, accuracy, and computational efficiency. Without the aid of an optimisation algorithm, the sensors are arranged in an ad-hoc configuration by the designer, which requires expertise or trial and error, which can be time-consuming or sacrificial if a sub-par arrangement is settled on with no mathematical quantitative guidance. Thus, the resulting localisation error is neither calculated nor considered mathematically. Here lies the need to optimise the camera network based on metrics that minimise degrading effects like resolution uncertainty and probabilistic occlusion even if the trajectory of the target is not known exactly beforehand.

5.1.3 Optimisation Approaches for Sensor Placement

Recent research has diverged into avenues of how to formulate the problem, when to conclude that an optimal solution has been found, and what consitutes success as a benchmark. Two primary classes of optimisation methods have emerged: heuristic algorithms and integer programming. The problem formulation typically depends of defining a specific objective function, sensor model, and workspace representation. Eventhough the search space is continuous it is typical of designers to discretise both the workspace and the possible camera locations to make the problem more computable. Depending on the application, available resources, and workspace constraints, different optimisation objectives have been prioritised in the literature. Broadly, these can be categorised into coverage-based objectives and accuracy-based objectives. In terms of coverage-based objectives, the goal is to maximise the visible area of the workspace, and sometimes going further to handle aspects of occlusion both static and dynamic and hence usually ensuring a two-cover coverage. This is particularly relevant in surveillance applications where ensuring that all regions are monitored is critical [2]. Conversely, accuracy-based objectives focus on minimising localisation error, which is crucial in motion capture and augmented reality applications where precise tracking is required and more than one camera arranged sufficiently non-parallel is needed to ensure triangulation is possible [3, 5]. Wu, Sharma, and Huang proposed another accuracy-based metric of placement which assessed how the limited camera intrinsics and quantisation lead to inaccurate measurements exacerbated by the placement of two cameras [8]. Many studies have proposed hybrid objectives that balance the trade-off of accuracy with coverage [4,

7]. Some works have also considered additional constraints such as budget limits and physical mounting options [3, 6, 9].

5.1.3.1 Optimisation Objectives

The configuration quality for 3D marker tracking and trinagulation is mostly affected by two main sources of error: marker visibility and triangulation accuracy. In terms of the application being Motion Capture (MoCap), a marker should not be occluded by any obstacle, including itself, the marker must be within the cameras field of view, vissible given the camera's resolution and range of capture. These are necessary to ensure consistent and continual tracking. Furthermore, at least two cameras should be arranged sufficiently non-parallel so that triangulation is well-conditioned. Sometimes, the easy answer is just to throw more cameras at the problem as that does give redundant views and ensures we can place camera pairs to cover the scene in a sufficently non-parallel format but in most research situationships, it is not prudent to over-exhaust the budget and hence it may be a desirable goal to asses the least amount of cameras required to achieve a level of accuracy/ coverage to minimise cost.

Provided below is a summary table of the different optimisation qualities found across papers in the literature (insert table with rows indication paper citation and application and columns of optimsation objectives)

5.1.4 Modelling the Problem

As mentioned earlier due to the large search space which is made so large becaus ethe problem is combinatorial (meaning additonal combinations of different aspects like position and roientation of each camera) which is NP-hard and increases expeonently with each additonal sensor added to the network it becomes necessary to make some simplifications as an exhaustive search especially for a room-sized scale search simply becomes intractable and computationally inefficent. This is often why the search-space is dicsretised both in the flight-space arena and in the possible mounting locations of the camera.

5.1.5 Summary

As seen there have been a plethora of proposed methodologies for optimising camera placement in the literature which are summarised in the timeline below where they have been broadly categorised as heuristic optimisation or binary integer programming and their objectives have also been listed based on thematic colour rings indicated in the key.

5.2 Theoretical Framework

5.2.1 Genetic Algorithm Terminology

5.2.2 Encoding Schemes

5.2.3 Selection Methods

5.2.4 Crossover Techniques

5.2.5 Convergence Analysis

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5.3.2.2 Dynamic Occlusion Cost Function

5.3.2.3 Combined Cost Function

5.3.3 Termination Criteria

5.4 Results and Discussion

5.4.1 Convergence Behaviour

5.4.2 Cost Function Evaluation

5.4.3 Parameter Sensitivity Analysis

5.4.4 Optimal Placement Configurations

Different Modalities Random Placement, ad hoc placement

5.4.5 System Limitations

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Conclusion and Future Work 6

6.1 Summary of Contributions

6.2 Key Findings

6.3 Limitations

includes scope constraints outlined in intro

6.4 Recommendations for Future Work

closed-loop robotic control Benchmark GA against other metaheuristic like PSO or SA to see if better solutions existan assess convergence performance

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